



6 WEEKS AI PROGRAM

Mind Map

A complete day by day breakdown of every module, every topic and exactly what you will learn across six weeks.

6 WEEKS	30 DAYS	250+ TOPICS	20+ PROJECTS
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Prepared for enrolled students

About This Program

This document is your complete map of the six week AI program. Every single day, module and topic in the curriculum is listed here with a plain description of what it covers, so you always know what is coming, what you are expected to build, and how each piece connects to the next.

The program is built on one principle: you learn by building. Concepts are never taught in isolation. Every day ends with something you have made, and most days end with an assignment that is reviewed. By the end of six weeks you will not just understand AI, you will have shipped automations, RAG systems, multi agent applications, a fine tuned model and your own small language model.

The Learning Journey

WEEK	FOCUS	WHAT YOU COVER
1	Foundations	AI fundamentals, Python, RPA, APIs, Streamlit and Gen AI basics
2	Prompting & RAG	Prompt engineering, JSON prompting, LangChain and Retrieval Augmented Generation
3	Automation	Knowledge graphs and complete AI automation with n8n
4	AI Agents I	Agents in n8n, then multi agent systems with CrewAI
5	AI Agents II	AutoGen framework and the production grade Hermes Agent
6	Model Building	Fine tuning, LoRA/QLoRA, Small Language Models and Claude Code

WEEK

01

Foundations - AI, Python, Automation and Backend

Week 1 builds the ground floor. You start with a clear mental model of what AI actually is, then pick up just enough Python to build real things, automate your desktop and browser, expose your work through APIs and interfaces, and finish with the fundamentals of Generative AI and LLMs.

THIS WEEK

Day 1	Introduction to AI (11 topics)
Day 2	Introduction to Python (17 topics)
Day 3	Introduction to RPA (13 topics)
Day 4	Introduction to API (Backend) (11 topics)
Day 5	Basics of Gen AI (11 topics)

Day 1

Introduction to AI

A complete orientation to the AI landscape. By the end of this day you will know where AI fits into your career, which skills actually matter, and your machine will be fully set up for the next six weeks.

#	TOPIC	WHAT IT COVERS
1	What is AI	The core definition of Artificial Intelligence, a short history of how it evolved, and everyday examples of AI already working around you.
2	AI vs ML vs DL vs Gen AI	The four layers explained clearly - how Machine Learning sits inside AI, Deep Learning inside ML, and Generative AI on top of Deep Learning, with the difference in purpose for each.
3	What is Gen AI	How generative models create new text, images, audio and code instead of just classifying data, and why this shift changed the industry.
4	Where to Apply AI	Identifying high value use cases across marketing, sales, operations, education, healthcare and content, plus a simple filter to judge whether a problem is worth solving with AI.
5	Domain Expert + Gen AI	The most valuable combination in the market today - why your existing industry knowledge multiplied by AI skill beats generic technical ability.
6	Five Skills in AI	The five capability areas this program covers end to end, and how each one stacks on the previous to make you job and business ready.
7	Opportunity in AI	Career paths, freelancing routes, agency models and product opportunities available right now, with a realistic view of what each demands.
8	Implementation Methods - Learn	How to learn effectively in this program - build while you learn, ship small projects daily, and convert every concept into working code.
9	Your Daily Task	The daily practice discipline: what to submit, how assignments are reviewed, and the habit loop that keeps you consistent for six weeks.
10	Platform Setup - VS Code, Python, GitHub	Full development environment setup on Windows, including the six week GitHub repository template where all your work will be versioned and submitted.
11	Installing Python and VS Code on MacBook	The same environment setup walked through step by step for macOS users, including terminal basics and PATH configuration.

Topics in this session: 11

Day 2

Introduction to Python

A fast but complete Python foundation designed for AI work. Every concept is taught with the end goal in mind - writing scripts, agents and applications, not passing exams.

#	TOPIC	WHAT IT COVERS
1	What is Programming	How instructions become software, what an interpreter does, and the mindset shift from user to builder.
2	Python and Uses	Why Python became the default language of AI, and the categories of work it handles - automation, data, web backends and machine learning.
3	Python Syntax	Indentation rules, statements, comments and the readable structure that makes Python different from other languages.
4	Python Virtual Environment	Creating isolated environments with venv, why dependency isolation matters, and how to manage requirements files per project.
5	How to Run Python in VS Code	Interpreter selection, running scripts, using the integrated terminal and the debugger basics.
6	Python Operators	Arithmetic, comparison, logical, assignment, membership and identity operators with practical examples.
7	Python Data Types	Strings, integers, floats, booleans and None - type conversion, mutability and common pitfalls.
8	Lists, Sets, Tuples, Dictionary	The four core collections, when to use each, and the operations you will use daily in AI code.
9	Python Control Flow	if, elif and else decision making, truthiness and writing clean conditional logic.
10	Nested If Statements	Handling multi level conditions, and refactoring deeply nested logic into readable code.
11	Python Loops Explained	for and while loops, range, break, continue, and iterating over collections and files.
12	Python Functions	Defining reusable functions, parameters, default values, return values, scope and docstrings.
13	Python Modules	Importing from the standard library, installing packages with pip, and structuring your own reusable modules.
14	Error Handling	try, except, else and finally, raising exceptions, and writing code that fails gracefully in production.
15	Python File Handling	Reading and writing text, CSV and JSON files, context managers, and safe file paths.
16	Python Recap	A consolidated revision of everything covered, highlighting the specific pieces you will reuse throughout the program.
17	Python Live Assignment	A guided hands on build where you apply the full day's learning in a single working script, reviewed live.

Topics in this session: 17

Day 3

Introduction to RPA

Robotic Process Automation gives you the power to make the computer do repetitive work for you. You will automate the desktop with PyAutoGUI and the browser with Playwright.

#	TOPIC	WHAT IT COVERS
1	What is RPA?	Robotic Process Automation explained - which tasks qualify, where it saves real hours, and how RPA differs from AI automation.
2	PyAutoGUI Basics	Installing and configuring PyAutoGUI, understanding screen coordinates, and safety controls like failsafe and pause.
3	PyAutoGUI Key Functionalities	The core function set for locating elements on screen, timing actions and building reliable automation flows.
4	PyAutoGUI Mouse Operation	Programmatic movement, clicks, double clicks, drag and drop and scrolling with precise positioning.
5	PyAutoGUI Keyboard Operation	Typing text, pressing keys, hotkey combinations and simulating full keyboard workflows.
6	PyAutoGUI Final Screenshot	Capturing the screen, region based screenshots and image recognition to locate buttons dynamically.
7	PyAutoGUI Demo Final	A complete end to end desktop automation built live, combining mouse, keyboard and screen detection.
8	PyAutoGUI Assignment	Build your own desktop automation for a repetitive task of your choice and submit it.
9	Playwright Basics	Browser automation fundamentals - installation, browser contexts, headless versus headed mode.
10	Playwright First Script	Your first working script: launch a browser, navigate, interact with a page and close cleanly.
11	Playwright Key Functions	Selectors, locators, waits, form filling, clicking, extracting content and handling multiple pages.
12	Playwright Demo	A full browser automation walkthrough on a real website, including data extraction.
13	Playwright Assignment	Automate a real browser workflow end to end and submit your script with output.

Topics in this session: 13

Day 4

Introduction to API (Backend)

Everything you build needs a way to talk to the outside world. This day covers APIs conceptually, then FastAPI for backends and Streamlit for fast user interfaces.

#	TOPIC	WHAT IT COVERS
1	Introduction to API	What an API is, the client server model, and why every AI tool you use is really an API call underneath.
2	API and its Types	REST, SOAP, GraphQL and WebSocket compared, with guidance on which you will actually encounter.
3	REST API and Methods	GET, POST, PUT, PATCH and DELETE, status codes, headers, request bodies and response structures.
4	Postman Installation	Setting up Postman, creating collections, sending requests and reading responses before you write any code.
5	What is FastAPI	The modern Python framework for building APIs - async support, automatic documentation and data validation with Pydantic.
6	FastAPI Demo	Building and running your first API with multiple endpoints, testing it through the auto generated docs.
7	Introduction to Streamlit	Turning Python scripts into shareable web apps without any frontend code.
8	Python Streamlit Components	Inputs, buttons, sliders, file uploaders, layout columns, sidebars and session state.
9	Build a Basic Streamlit App	Constructing a complete interactive app from scratch, live.
10	Streamlit Assignment Explained	A full walkthrough of the assignment brief, expected features and evaluation criteria.
11	Streamlit Assignment Details and Resources	Reference material, starter code and documentation links to complete the assignment independently.

Topics in this session: 11

Day 5

Basics of Gen AI

The LLM landscape decoded. Open source versus closed source, how to get API access, how to read benchmarks, and how to run models locally on your own machine.

#	TOPIC	WHAT IT COVERS
1	LLM and its Types	What a Large Language Model is, how it is trained, and the families of models available - base, instruct, reasoning and multimodal.
2	Open Source vs Cloud LLM	The trade offs between self hosted open models and hosted cloud models across cost, privacy, speed and capability.
3	Closed Source LLM	The major commercial providers, their model tiers and the strengths of each for different tasks.
4	How to Purchase API Keys for Closed LLMs	Account creation, billing setup, generating keys and storing them securely with environment variables.

#	TOPIC	WHAT IT COVERS
5	LLM Benchmarks	Reading leaderboards and benchmark scores critically, and choosing a model based on your actual use case rather than hype.
6	Essential AI Tools	The working toolkit used throughout this program and in professional AI practice.
7	Hugging Face Spaces	Exploring, using and deploying models and demos on the largest open AI community platform.
8	Pulling Open Source Models from Hugging Face	Downloading model weights, understanding model cards, licences and hardware requirements.
9	Running Open Source Models with Ollama	Installing Ollama, pulling models, running inference locally and calling them from Python.
10	Ollama Assignment	Run an open source model locally and build a small application on top of it.
11	Weekend Assignment - Quiz App	A capstone for Week 1: build a quiz application using Streamlit for the interface and FastAPI for the backend, tying together everything learned.

Topics in this session: 11

WEEK

02

Prompting, LangChain and RAG

Week 2 turns you from an AI user into an AI engineer. You master structured prompting, learn the LangChain ecosystem component by component, and build Retrieval Augmented Generation systems that ground models in your own data.

THIS WEEK

Day 1	Prompt Engineering (18 topics)
Day 2	JSON Prompting (8 topics)
Day 3	LangChain Ecosystem (10 topics)
Day 4	Introduction to RAG (9 topics)
Day 5	LangChain Advanced (5 topics)

Day 1

Prompt Engineering

The single highest leverage skill in Gen AI. You learn a repeatable framework, then advanced techniques used in production systems.

#	TOPIC	WHAT IT COVERS
1	My Story on Prompting - Manoj	A personal account of how deliberate prompting changed real business outcomes, and the lessons learned the hard way.
2	What is Prompt Engineering	The discipline of designing inputs that reliably produce the output you want, and why it is engineering rather than guesswork.
3	System Prompt vs User Prompt	How the two layers differ, what belongs in each, and how system prompts control behaviour across an entire application.
4	Basic Prompting Formula - RTCFR	The core framework taught in this program, giving you a repeatable structure for every prompt you write.
5	R - Role	Assigning the model an identity and expertise level, and how this shifts tone, depth and accuracy.
6	T - Task	Writing unambiguous task instructions with clear scope, constraints and success criteria.
7	C - Context	Supplying background, audience, data and situational detail so the model is not guessing.
8	F - Few Shot	Using worked examples to teach pattern and format, and how many examples are actually needed.
9	R - Response	Specifying output format, length, structure and tone so the result is usable without editing.
10	RTCFR Framework Demo	The full framework applied live to real prompts, showing the before and after difference.
11	RTCFR Prompt PDF Guide	A downloadable reference guide with templates and examples you can reuse immediately.
12	Prompt Tricks and Tips	Practical techniques that consistently improve output quality across models.
13	Reverse Prompting	Having the model interrogate you to build the ideal prompt, useful when you know the goal but not the wording.
14	Prompt Chaining	Breaking complex work into a sequence of prompts where each output feeds the next step.
15	Advanced Prompting - CoT, ToT, ReAct	Chain of Thought for step by step reasoning, Tree of Thoughts for exploring multiple paths, and ReAct for interleaving reasoning with tool use.
16	Keywords of Prompting	A curated vocabulary of instruction words that reliably steer model behaviour, supplied as a PDF reference.

#	TOPIC	WHAT IT COVERS
17	Prompt Injection	How malicious inputs hijack model instructions, real attack patterns, and the defences you must build into applications.
18	Prompting Exam	A formal assessment to validate your prompting skill before moving into framework based development.

Topics in this session: 18

Day 2 JSON Prompting

Structured prompting for structured output. JSON and Markdown formats that make AI output machine readable and production ready, applied to text, image and video generation.

#	TOPIC	WHAT IT COVERS
1	Introduction to JSON Prompting	Why structured prompts produce more consistent, parseable results than free form text.
2	SCALE Framework	A structured framework for building detailed JSON prompts with controllable parameters.
3	Image Generation Using SCALE	Applying the framework to image models for precise control over subject, style, lighting, composition and mood.
4	What is .md?	Markdown syntax, structure and why it has become the standard format for AI readable documents.
5	Why We Need MD Files for AI Interaction	How Markdown files serve as persistent instructions, context and configuration for AI systems and agents.
6	Sample .md File	A worked example file you can use as a template for your own AI instruction documents.
7	JSON Prompting - Video Generation	Structured prompting applied to video models, controlling scene, camera movement, duration and continuity.
8	JSON Prompting Assignment	Produce structured prompts for text, image and video generation and submit the outputs.

Topics in this session: 8

Day 3 LangChain Ecosystem

The framework that connects LLMs to data, tools and memory. Each component is taught individually so you understand what the abstractions are actually doing.

#	TOPIC	WHAT IT COVERS
1	Intro to LangChain	What problem LangChain solves, and why raw API calls stop being enough as applications grow.

#	TOPIC	WHAT IT COVERS
2	LangChain Ecosystem	The full family - LangChain, LangGraph, LangSmith and LangServe - and where each fits.
3	LangChain Components	The building blocks you will assemble: models, prompts, chains, retrievers, memory and output parsers.
4	Data Ingestion	Document loaders for PDFs, web pages, CSVs, databases and APIs to get your data into the pipeline.
5	Text Splitter	Chunking strategies, chunk size and overlap, and how splitting decisions directly affect retrieval quality.
6	Embeddings	Converting text into vectors, how semantic similarity works, and choosing an embedding model.
7	Vector Store	Storing and indexing embeddings, similarity search, and comparing the popular vector database options.
8	Retriever	Fetching the most relevant chunks for a query, with search types and filtering.
9	Chain	Composing components into sequences using LCEL, including branching and parallel execution.
10	LangSmith	Tracing, debugging, monitoring and evaluating your LLM applications with full visibility into every step.

Topics in this session: 10

Day 4 Introduction to RAG

Retrieval Augmented Generation lets a model answer from your documents instead of only its training data. You cover the standard pipeline and the advanced variants used in production.

#	TOPIC	WHAT IT COVERS
1	Introduction to RAG	The full RAG pipeline from ingestion to grounded answer, and why it beats fine tuning for knowledge tasks.
2	RAG Types	The landscape of RAG architectures and the specific problems each variant solves.
3	Self RAG	Letting the model decide when retrieval is needed and critique its own retrieved context.
4	Corrective RAG	Detecting poor retrievals and correcting them with fallback searches before generating an answer.
5	Fusion RAG	Generating multiple query variations and fusing results for significantly better recall.
6	BM25 and Reranking	Combining keyword search with vector search, then reranking results to put the best context first.

#	TOPIC	WHAT IT COVERS
7	RAG Demo	A complete working RAG system built live over a real document set.
8	RAG Demo with LangSmith	Instrumenting the same system with tracing to see retrieval quality, latency and cost at every step.
9	RAG Live Assignment	Build your own RAG application over your own documents and demonstrate it.

Topics in this session: 9

Day 5

LangChain Advanced

Production concerns - integrating multiple models, managing conversation state, adding safety controls and measuring quality.

#	TOPIC	WHAT IT COVERS
1	Model Integration	Connecting different providers and open source models through a single interface, and swapping models without rewriting code.
2	Messages	System, human, AI and tool messages, how conversation history is structured and passed.
3	Guardrails Basics	Input and output validation, content filtering, and preventing your application from producing unsafe or off topic responses.
4	Evaluation Basics	Defining quality metrics, building test datasets and systematically measuring whether changes actually improve your application.
5	Building a Basic Gen AI Application	Assembling everything from Week 2 into one complete, working application.

Topics in this session: 5

WEEK

03

Knowledge Graphs and AI Automation with n8n

Week 3 expands retrieval into knowledge graphs, then moves into workflow automation. You learn n8n end to end - self hosting, nodes, ETL pipelines, authentication and real API integrations with Gmail, Telegram and Slack.

THIS WEEK

Day 1	Knowledge Graph (4 topics)
Day 2	n8n Fundamentals (5 topics)
Day 3	n8n Nodes and ETL (8 topics)
Day 4	n8n API Integrations (8 topics)
Day 5	n8n Concepts and Offline Preparation (5 topics)

Day 1

Knowledge Graph

When relationships between entities matter more than text similarity, graph based retrieval outperforms vector search. This day covers the concept and a full working implementation.

#	TOPIC	WHAT IT COVERS
1	Knowledge Graph - Intro	Nodes, edges and properties - representing entities and their relationships, and where graphs beat plain vector retrieval.
2	Knowledge Graph RAG Pipeline	Extracting entities and relationships from documents, building the graph, and querying it to answer multi hop questions.
3	KG RAG Demo with Neo4j Aura DB and NetworkX	A hands on build using Neo4j Aura for a hosted graph database and NetworkX for in memory graph work.
4	Knowledge Graph Assignment	Construct a knowledge graph from your own dataset and query it through a RAG pipeline.

Topics in this session: 4

Day 2

n8n Fundamentals

The starting point for AI automation - understanding what separates plain automation from AI automation, and getting n8n running on your own infrastructure.

#	TOPIC	WHAT IT COVERS
1	Automation vs AI Automation	The difference between rule based workflows and workflows where a model makes decisions, and when each is appropriate.
2	AI Automation Types	The categories of AI automation - classification, extraction, generation, routing and agentic - with examples of each.
3	Platforms for AI Automation	Comparing n8n, Make, Zapier and code first approaches on cost, flexibility, data control and scale.
4	n8n Installation Types	Cloud, npm, Docker and desktop options compared, with guidance on choosing for your situation.
5	n8n Self Host	Setting up your own n8n instance with full control over data, credentials and custom nodes.

Topics in this session: 5

Day 3

n8n Nodes and ETL

Working fluently inside the n8n editor, understanding every node type, and building real ETL pipelines both visually and in Python.

#	TOPIC	WHAT IT COVERS
1	n8n Editor Tour	The canvas, node panel, execution view, logs and settings - navigating the interface confidently.
2	What is a Node and Types of Nodes	Triggers, actions, logic and transform nodes, and how data flows between them as items.
3	Node Types Full Overview	A complete walkthrough of the node categories available and what each is used for.
4	HTTP vs Webhook Node	Making outbound requests versus receiving inbound calls, and how to combine both in one workflow.
5	ETL AI Automation in n8n	Extract, Transform, Load pipelines built visually, with AI in the transform step.
6	ETL AI Automation in Python	The same pipeline written in code, so you understand exactly what the visual tool is doing.
7	ETL Demo in n8n	A complete pipeline built live from a real data source through to a destination system.
8	ETL AI Automation Assignment	Design and build your own ETL automation and submit the working workflow.

Topics in this session: 8

Day 4

n8n API Integrations

Connecting n8n to the real services your work depends on, with a proper understanding of authentication and access control.

#	TOPIC	WHAT IT COVERS
1	Authentication vs Authorization	Who you are versus what you are allowed to do, and why confusing the two causes security failures.
2	HTTP Node in n8n - Quick Overview	Configuring methods, headers, query parameters, body formats and pagination.
3	API Key, Token Access and OAuth 2.0	The three common auth mechanisms, the OAuth flow step by step, and securely storing credentials.
4	API AI Automation - Gmail	Building an automation that reads, classifies and responds to email intelligently.
5	API AI Automation - Telegram	Creating a Telegram bot workflow that receives messages and replies using AI.
6	API AI Automation - Slack and Custom GPT Workflow Builder	Slack integration for team automation, plus using a Custom GPT to generate n8n workflows from a description.



#	TOPIC	WHAT IT COVERS
7	Claude + n8n Connector	The simple method for connecting Claude directly to your n8n instance.
8	API AI Automation in Python	Implementing the same integrations in code for full control and portability.

Topics in this session: 8

Day 5

n8n Concepts and Offline Preparation

Advanced n8n patterns and the Docker setup required for everything that follows in Weeks 4 to 6.

#	TOPIC	WHAT IT COVERS
1	Agenda	What this session covers and how it prepares you for the agent weeks ahead.
2	UI Automation Connected with n8n	Bridging desktop RPA from Week 1 with cloud workflows so automations span both worlds.
3	Web Scraping Automation	Extracting structured data from websites on a schedule, handling pagination and anti bot considerations.
4	Docker Installation - Windows	Installing Docker Desktop on Windows, WSL 2 configuration and verifying the setup.
5	Docker Installation - MacBook	The same installation walked through for macOS, including Apple Silicon considerations.

Topics in this session: 5

WEEK

04

AI Agents - n8n and CrewAI

Week 4 is where AI stops answering and starts acting. You build agents in n8n with tools, memory and human oversight, then move into CrewAI to orchestrate teams of specialised agents collaborating on multi step work.

THIS WEEK

Day 1	AI Agents - Basics (6 topics)
Day 2	AI Agents - n8n Advanced (5 topics)
Day 3	Introduction to CrewAI (10 topics)
Day 4	CrewAI Core Concepts (10 topics)
Day 5	CrewAI Applied (7 topics)

Day 1

AI Agents - Basics

The architecture of an agent, then progressively harder builds - from a single tool Gmail agent to a multi tool personal assistant.

#	TOPIC	WHAT IT COVERS
1	What is an AI Agent?	The difference between a chatbot and an agent - goals, planning, tool use and autonomous action loops.
2	AI Agent Architecture	The core loop of perception, reasoning, action and observation, plus the role of memory and tool definitions.
3	Basic AI Agent - Gmail	Your first working agent that reads, understands and acts on email autonomously.
4	Amazon Research Agent - n8n Demo	An agent that researches products, extracts structured data and compiles findings into a report.
5	Multi Tool AI Agents	Giving one agent access to several tools and letting it choose which to use for each step.
6	Build Jarvis in n8n	A personal assistant agent combining multiple tools, memory and natural conversation into one system.

Topics in this session: 6

Day 2

AI Agents - n8n Advanced

Model Context Protocol, multi agent workflows, human approval gates and shipping an agent as a real web application.

#	TOPIC	WHAT IT COVERS
1	MCP Implementation in n8n	Connecting Model Context Protocol servers to give agents standardised access to external tools and data.
2	MCP at n8n Instance Level	Configuring MCP across your whole instance so every workflow can share the same tool set.
3	Multi AI Agent Workflow	Multiple specialised agents passing work between each other, with routing and coordination logic.
4	Human in the Loop	Inserting approval checkpoints so a person reviews critical actions before the agent proceeds.
5	Build a Web App with n8n	Exposing your agent workflow as a usable web application with a real interface.

Topics in this session: 5

Day 3

Introduction to CrewAI

Moving from visual workflows into a code first agent framework. Covers the prerequisites, then CrewAI's core building blocks of agents and tasks.

#	TOPIC	WHAT IT COVERS
1	Introduction to AI Agents and Agentic AI	A deeper conceptual foundation - autonomy levels, agentic patterns and where the field is heading.
2	Introduction to Pydantic Framework	Data models, type validation and structured outputs - the backbone of reliable agent inputs and outputs.
3	Introduction to Python Async Functions	async and await, concurrency versus parallelism, and why agent frameworks depend on async execution.
4	What is CrewAI	The framework's philosophy of role based agents collaborating like a real team.
5	CrewAI Architecture and Core Concepts	Agents, Tasks, Crews, Processes and Tools - how the pieces connect.
6	Installing and Setting Up CrewAI	Environment setup, project structure, configuration files and running your first crew.
7	LLM Integration in CrewAI	Connecting cloud and local models, assigning different models to different agents, and controlling cost.
8	Understanding Agents	Agent configuration, capabilities, delegation settings and execution limits.
9	Agent Roles, Goals and Backstory	Writing the three definition fields that shape how an agent thinks and behaves, with examples of good and poor definitions.
10	Tasks in CrewAI	Task descriptions, expected outputs, agent assignment, context passing and output files.

Topics in this session: 10

Day 4

CrewAI Core Concepts

Building real multi agent systems - collaboration patterns, custom tools, document processing, memory and knowledge sources.

#	TOPIC	WHAT IT COVERS
1	Creating Multi Step Workflows	Chaining tasks so each agent's output becomes the next agent's input, with dependency handling.
2	Crew and Agent Collaboration	How agents delegate, ask each other questions and combine their work into a single result.

#	TOPIC	WHAT IT COVERS
3	Process Types	Sequential versus hierarchical execution, with a manager agent coordinating work in hierarchical mode.
4	Tools in CrewAI	The built in tool library and how tool calling actually works under the hood.
5	Integrating Custom Python Tools	Writing your own tools so agents can call any function, API or internal system you have.
6	Web Search Tools	Giving agents live internet access for research, with result handling and source tracking.
7	File and Document Processing Tools	Reading, parsing and writing PDFs, spreadsheets and text files from within an agent workflow.
8	Memory in CrewAI	How agents retain context within and across runs, and the memory types available.
9	Short Term vs Long Term Memory	The distinction between working context and persistent knowledge, and configuring each correctly.
10	Knowledge Source and RAG	Attaching your own document knowledge base to agents so they answer from your data.

Topics in this session: 10

Day 5

CrewAI Applied

Vector databases, human oversight, debugging and two complete project builds that bring the week together.

#	TOPIC	WHAT IT COVERS
1	Vector Database Integration	Connecting a persistent vector store to your crew for scalable, long lived retrieval.
2	Human in the Loop	Adding review and approval points inside code based agent workflows.
3	Building a Research Assistant Crew	A complete multi agent system that researches a topic, verifies findings and produces a structured report.
4	Logging and Debugging Agents	Verbose output, execution traces and systematic techniques for diagnosing why an agent behaved unexpectedly.
5	What is MCP	Model Context Protocol explained properly - the standard for connecting agents to tools and data sources.
6	Blog Team App	A multi agent content team with researcher, writer and editor agents producing publishable output.
7	Live Assignment Building	Build and present your own CrewAI multi agent application.

Topics in this session: 7

WEEK

05

AutoGen and the Hermes Agent

Week 5 covers a second major agent framework and a production agent system. You build agents and teams in AutoGen with tools, memory and state management, then go deep into the Hermes Agent - its architecture, data flow, core files and complete setup.

THIS WEEK

Day 1	Introduction to AutoGen (8 topics)
Day 2	AutoGen Framework (6 topics)
Day 3	AutoGen Advanced and Projects (7 topics)
Day 4	Hermes Agent - Architecture (8 topics)
Day 5	Hermes Agent - Configuration (8 topics)

Day 1

Introduction to AutoGen

Microsoft's multi agent conversation framework. You configure it, build your first agent, and extend it with tools.

#	TOPIC	WHAT IT COVERS
1	Intro to AutoGen Framework	The conversational multi agent model, how AutoGen differs from CrewAI, and when to choose each.
2	Configuring the LLM	Model client setup, provider configuration, parameters and cost controls.
3	First Agent in AutoGen	Creating and running an assistant agent end to end.
4	Prompt Engineering in AutoGen	Applying Week 2 prompting skills to agent system messages for reliable behaviour.
5	Messages	The message types that flow between agents, and how conversation history is managed.
6	Observing Agents	Streaming output, inspecting decisions and following an agent's reasoning in real time.
7	Tools in Agents	Registering tools, function calling and handling tool results inside the conversation loop.
8	Custom Tools with Python	Writing your own Python tools with proper schemas so agents can call your code reliably.

Topics in this session: 8

Day 2

AutoGen Framework

Search, memory, and building teams of agents that work together with clear termination rules.

#	TOPIC	WHAT IT COVERS
1	Web Search Integration	Connecting live search so agents work with current information rather than training data alone.
2	Memory in AutoGen	Persisting context across turns and sessions, and the memory options available.
3	Teams in AutoGen	Group chats and team structures where multiple agents participate in one conversation.
4	Termination in AutoGen	Stop conditions, max turns and completion signals - preventing agents from looping forever.
5	Multi Agent Team Building	Designing a team with complementary roles and a clear handoff pattern.

#	TOPIC	WHAT IT COVERS
6	Team Operation	Running, monitoring and tuning a live multi agent team.

Topics in this session: 6

Day 3 AutoGen Advanced and Projects

State management, observability, human oversight and a complete project build with MCP integration.

#	TOPIC	WHAT IT COVERS
1	Managing States	Saving, loading and resuming agent and team state so long running work survives restarts.
2	Observing Team Agents	Monitoring multi agent conversations to understand and improve collective behaviour.
3	Human in the Loop	User proxy agents and approval gates for human controlled execution.
4	Overview of Documentation	Navigating the official docs so you can keep learning independently after the program.
5	AI Interviewer Project	Building a complete AI interviewer that asks questions, evaluates answers and produces feedback.
6	Connecting MCP with Agents	Wiring MCP servers into AutoGen agents for standardised tool access.
7	Live Assignment Building	Build and present your own AutoGen multi agent application.

Topics in this session: 7

Day 4 Hermes Agent - Architecture

A complete production agent system studied from the inside out - what it is, how it is structured, how data moves through it and what it costs to run.

#	TOPIC	WHAT IT COVERS
1	What is Hermes Agent	An overview of the system, what it was built to do and the problems it solves.
2	What is Inside Hermes Agent	The components that make up the system and how they fit together.
3	Hermes Agent Documentation	Navigating the documentation to find configuration, capabilities and reference material.
4	Hermes Agent Architecture	The full architectural design - layers, modules and the reasoning behind the structure.

#	TOPIC	WHAT IT COVERS
5	Hermes Agent Data Flow	Tracing a request from input through processing to output, including where state is stored.
6	Hermes Agent Directory Structure	The folder layout explained file by file so you can navigate and extend the codebase.
7	Hermes Agent Setup Cost	A realistic breakdown of infrastructure, model and operational costs before you commit.
8	Hermes Agent Setup and Installation	Step by step installation, dependencies, configuration and first successful run.

Topics in this session: 8

Day 5 Hermes Agent - Configuration

The configuration files that define agent identity, user context, memory and capability - then putting it to work.

#	TOPIC	WHAT IT COVERS
1	Hermes Agent - 4 Core Files	The four files that control the entire system, and how editing them changes behaviour.
2	Hermes Agent - Skills and Connectors	Adding capabilities through skills and connecting external services through connectors.
3	Hermes Agent - Setup Guide	A consolidated, followable guide for a clean setup from zero.
4	Soul.md	Defining the agent's identity, personality, values and operating principles.
5	User.md	Storing who the user is, their preferences and context so the agent personalises its work.
6	Memory.md	How persistent memory is written, structured and recalled across sessions.
7	Agents.md	Defining sub agents, their roles and how work is delegated between them.
8	Hermes Agent - Create Content	Using the fully configured agent to produce real content end to end.

Topics in this session: 8

WEEK

06

Fine Tuning, Small Language Models and Claude Code

The final week takes you under the hood. You fine tune your own LLM with LoRA and QLoRA, build a Small Language Model through knowledge distillation, serve both as APIs, and finish by setting up Claude Code for agentic development.

THIS WEEK

Day 1	Introduction to Fine Tuning (8 topics)
Day 2	Fine Tuning Implementation (8 topics)
Day 3	SLM Building - Foundations (5 topics)
Day 4	SLM Building - Training and Serving (5 topics)
Day 5	Claude Code Setup (4 topics)

Day 1

Introduction to Fine Tuning

When to fine tune, what hardware you need, how to choose a base model, and how to build the dataset that determines your results.

#	TOPIC	WHAT IT COVERS
1	Introduction to Fine Tuning	What fine tuning actually changes inside a model, and the outcomes it can and cannot deliver.
2	Prompt Engineering vs RAG vs Fine Tuning	A decision framework for choosing the right approach based on cost, data and the problem type.
3	LLM Training Pipeline	Pre training, supervised fine tuning and alignment stages, and where your work fits in that pipeline.
4	Types of Fine Tuning	Full fine tuning, parameter efficient methods, instruction tuning and domain adaptation compared.
5	Hardware and Environment Setup	GPU requirements, VRAM planning, cloud options and installing the training stack.
6	Choosing the Right Base Model	Matching model size, licence, architecture and language capability to your use case and budget.
7	Understanding Datasets for Fine Tuning	Dataset formats, instruction response structure, quality signals and how much data you actually need.
8	Creating and Preparing Your Own Dataset	Collecting, cleaning, formatting and splitting your data into train and validation sets.

Topics in this session: 8

Day 2

Fine Tuning Implementation

The full hands on cycle - train, monitor, evaluate, merge, run inference and ship it as an API.

#	TOPIC	WHAT IT COVERS
1	Tokenization and Embeddings	How text becomes tokens and tokens become vectors, plus tokenizer choice and special tokens.
2	LoRA and QLoRA Configuration	Low Rank Adaptation explained, rank and alpha settings, target modules and quantised training for limited hardware.
3	Fine Tune Your First LLM	A complete hands on training run from prepared dataset to trained adapter.
4	Monitoring and Overfitting	Reading loss curves, spotting overfitting early, and applying early stopping and regularisation.

#	TOPIC	WHAT IT COVERS
5	Evaluating the Model	Quantitative metrics and qualitative testing to prove the fine tuned model is genuinely better.
6	Merge LoRA and Saving the Model	Merging adapter weights into the base model and saving in a deployable format.
7	Running Model Inference	Loading your model, generation parameters, batching and optimising response speed.
8	Building the Model as an API with FastAPI	Wrapping your fine tuned model in a production API with proper request handling.

Topics in this session: 8

Day 3 SLM Building - Foundations

Creating a small, fast, cheap model that inherits capability from a larger one through knowledge distillation.

#	TOPIC	WHAT IT COVERS
1	Introduction to SLM Building	Why Small Language Models matter - speed, cost, privacy and edge deployment.
2	Teacher and Student Model	The distillation relationship, selecting a capable teacher and sizing the student appropriately.
3	Knowledge Distillation Concepts	Soft targets, temperature, distillation loss and how knowledge transfers between models.
4	Preparing Distillation Data	Generating and curating the teacher outputs that the student will learn from.
5	Building the Student by Dropping Layers	Layer pruning to construct a smaller architecture while preserving as much capability as possible.

Topics in this session: 5

Day 4 SLM Building - Training and Serving

Running the distillation loop, proving the student works, and deploying it.

#	TOPIC	WHAT IT COVERS
1	Distillation Training Loop	Implementing the training loop, loss functions, hyperparameters and monitoring.
2	Evaluating the Student Model	Benchmarking the student against the teacher on quality, latency and cost.

#	TOPIC	WHAT IT COVERS
3	Save and Run Your Small Model	Exporting, loading and running inference on your own compact model.
4	Serving with FastAPI	Deploying the SLM as an API endpoint ready for real applications.
5	SLM Building Overview	A consolidated recap of the full pipeline from teacher selection to deployed student.

Topics in this session: 5

Day 5

Claude Code Setup

The closing session - moving from AI assisted coding to agentic development, with full environment setup on both platforms.

#	TOPIC	WHAT IT COVERS
1	AI Assisted vs Agentic Assisted	The difference between autocomplete style help and an agent that plans, edits files and runs commands on your behalf.
2	Claude Ecosystem and Subscription	The available models, interfaces and plans, and choosing what fits your workflow and budget.
3	Claude Installation - Mac	Complete installation and configuration on macOS, including authentication and first project.
4	Claude Installation - Windows	The same setup walked through for Windows users end to end.

Topics in this session: 4

What You Walk Away With

Six weeks is short. What makes it work is that nothing here is theory for its own sake. Every module exists because it is something you will actually use.

▶ **Build with Python**

A working command of Python for automation, backends and AI applications.

▶ **Automate anything repetitive**

Desktop RPA, browser automation and full workflow automation in n8n.

▶ **Engineer prompts properly**

A repeatable framework plus advanced reasoning techniques and injection defence.

▶ **Ground AI in your own data**

Production RAG systems, vector stores, reranking and knowledge graphs.

▶ **Ship AI agents**

Single and multi agent systems across n8n, CrewAI, AutoGen and the Hermes Agent.

▶ **Train your own models**

Fine tuned LLMs with LoRA and QLoRA, and a distilled Small Language Model.

▶ **Serve it to real users**

FastAPI backends and Streamlit interfaces for everything you build.

▶ **Work agentially**

Claude Code set up and integrated into your everyday development workflow.

One last thing.

The students who get the most out of this program are not the ones with the strongest technical background. They are the ones who show up daily, submit every assignment, and keep building even when the code breaks. Treat this document as your commitment list. Six weeks from now, the difference will be visible in what you have built.



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